



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



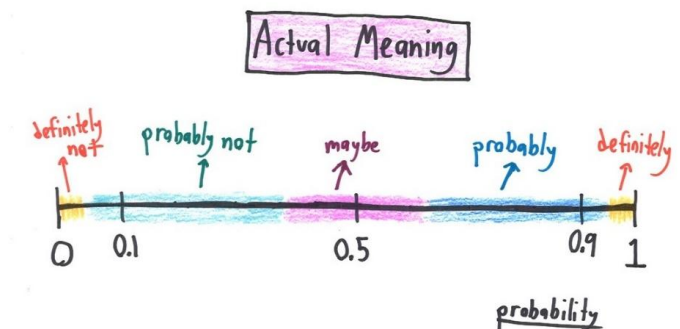


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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Week 13 - Lecture No 24 – November 10, 2025



Announcements!

1. All lectures and grades posted.
2. Assignment No 02 uploaded – **due date November 10, 2025 (today)**
3. Assignment 3 to be uploaded soon
4. **Quiz No 09 – November 24, 2025 (from Unit 5)**

Agenda for today

1. Unit 5 – Continuous Random Variables
 - Conditioning a r.v. on an event
 - Conditional PDF
 - Conditional expectation and the expected value rule
 -  – Exponential PDF: memorylessness
 -  – Total probability and expectation theorems
 -  – Mixed distributions

Unit 5
Continuous Random Variables

Memorylessness of the exponential PDF

- Bulb lifetime T : exponential (λ)
- Do you prefer a used or new “exponential light bulb”? *Probabilistically identical!*

$$P(T > x) = e^{-\lambda x}, \text{ for } x \geq 0$$

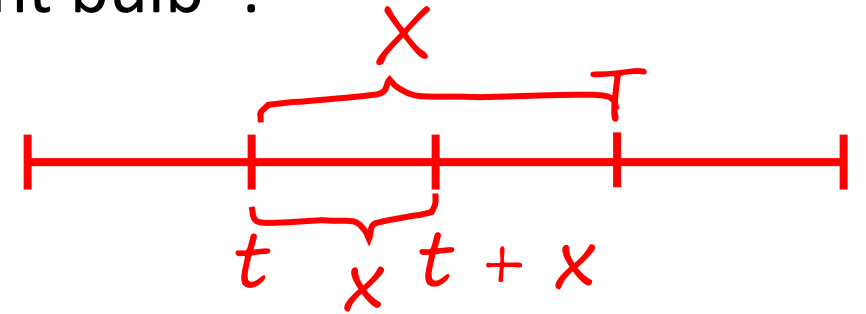
- We are told that $T > t$

- r.v. X : remaining lifetime $= T - t$

$$P(X > x \mid T > t) = e^{-\lambda x}, \text{ for } x \geq 0$$

$$= \frac{P(T - t > x, T > t)}{P(T > t)} = \frac{P(T > t + x, T > t)}{P(T > t)} = \frac{P(T > t + x)}{P(T > t)}$$

$$= \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$



Memorylessness of the exponential PDF

$$f_T(x) = \lambda e^{-\lambda x}, \text{ for } x \geq 0$$

$$P(0 \leq T \leq \delta) \approx \cancel{f_T(0)} * \delta = \lambda \delta$$

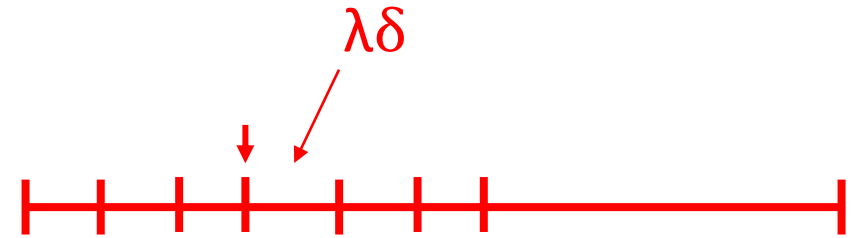
$$P(t \leq T \leq t + \delta \mid T > t) = P(0 \leq T \leq \delta) \approx \lambda \delta$$

similar to an independent coin flip,

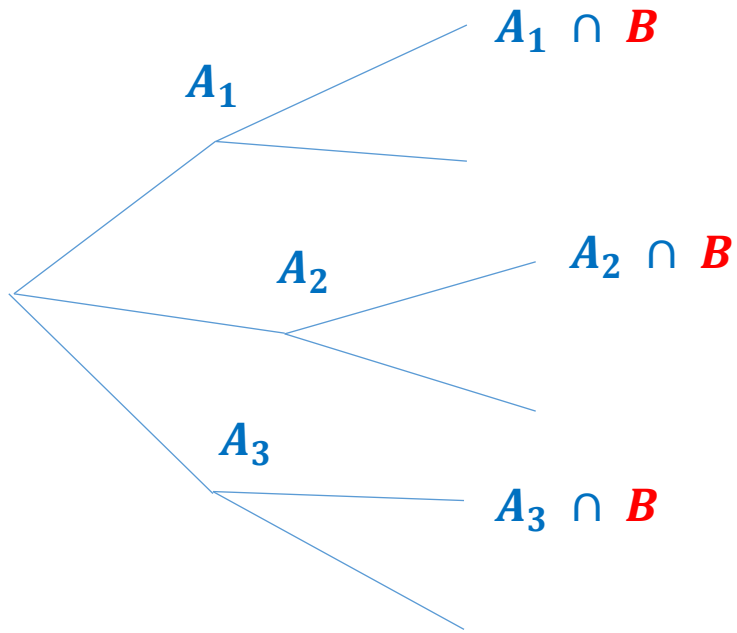
Every δ time steps,

With $P(\text{success}) \approx \lambda \delta$

Can human life be modelled like this?



Total probability and expectation theorems



$$P(B) = P(A_1) P(B | A_1) + \dots + P(A_n) P(B | A_n)$$

$$p_X(x) = P(A_1) p_{X|A_1}(x) + \dots + P(A_n) p_{X|A_n}(x)$$

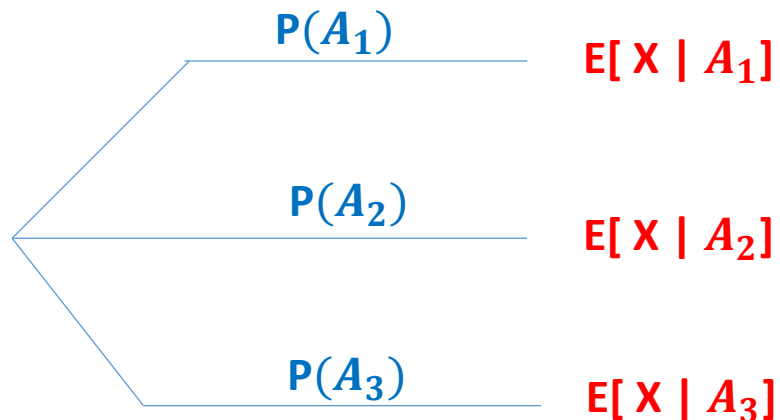
$$F_X(x) = P(X \leq x) = P(A_1) P(X \leq x | A_1) + \dots$$

$$= P(A_1) F_{X|A_1}(x) + \dots$$

$$f_X(x) = P(A_1) f_{X|A_1}(x) + \dots + P(A_n) f_{X|A_n}(x)$$

$$\int x \cdot f_X(x) dx = P(A_1) \int x \cdot f_{X|A_1}(x) dx + \dots$$

$$E[X] = P(A_1) E[X|A_1] + \dots + P(A_n) E[X|A_n]$$



Example

Sameena goes to the supermarket shortly, with probability $1/3$, at a time uniformly distributed between 0 and 2 hours from now; or with probability $2/3$, later in the day, at a time uniformly distributed between 6 to 8 hours from now. Find the pdf $f_X(x)$, where RV X is the time of departure to the supermarket. Also, calculate $E[X]$.

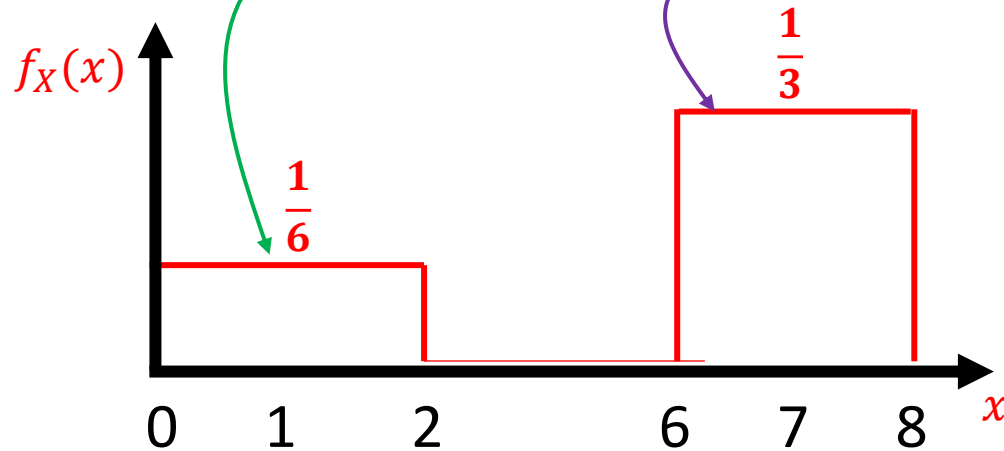
$$f_{X|A_1} \sim U[0, 2]$$

$$f_{X|A_2} \sim U[6, 8]$$

$$P(A_1) = \frac{1}{3}$$

$$P(A_2) = \frac{2}{3}$$

$$f_X(x) = P(A_1)f_{X|A_1}(x) + \dots + P(A_n)f_{X|A_n}(x)$$



$$E[X] = P(A_1) \cdot E[X|A_1] + \dots + P(A_n) \cdot E[X|A_n]$$

$$\frac{1}{3} * 1 + \frac{2}{3} * 7$$

Mixed Distributions

You play lottery and you have 1 dollar.

$$X = \begin{cases} 1, & \text{with probability } \frac{1}{2} \\ \text{uniform on } [0, 2], & \text{with probability } \frac{1}{2} \end{cases}$$

Is X Discrete? **No**

Is X Continuous? **No**

Y Discrete
 Z Continuous

$$X = \begin{cases} Y, & \text{with probability } (1 - p) \\ Z, & \text{with probability } p \end{cases}$$

$$P(X=1) = \frac{1}{2}$$

X is mixed

$$\rightarrow F_X(x) = P(X \leq x)$$

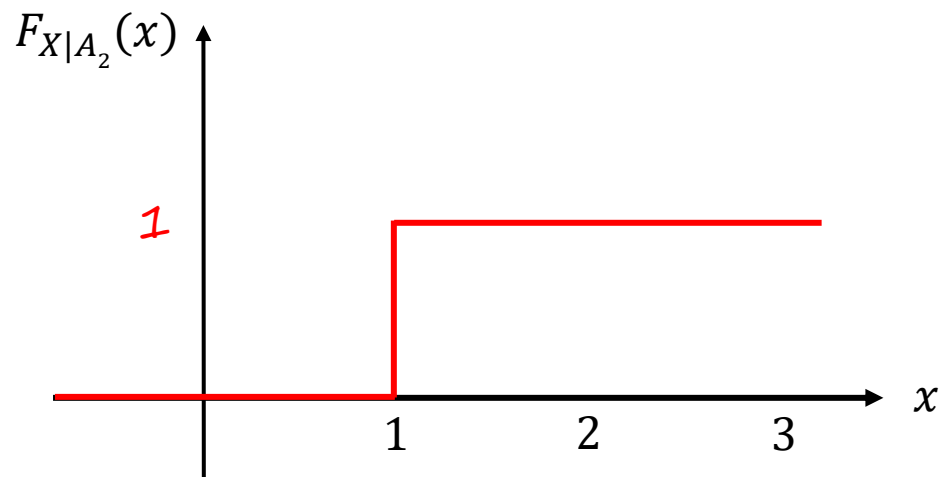
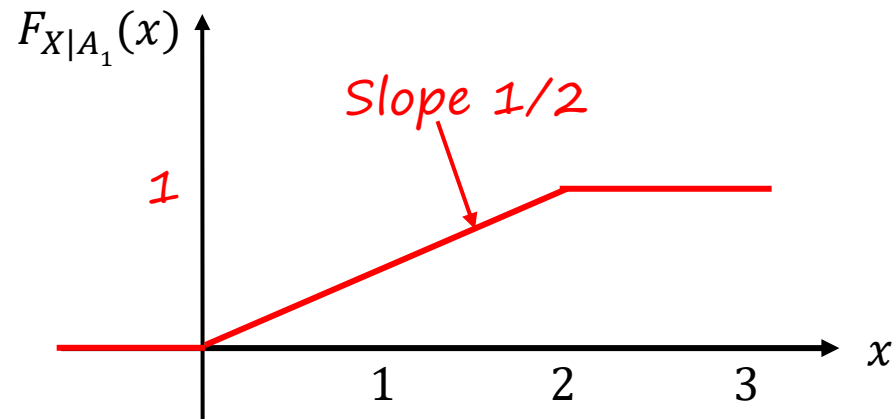
$$\rightarrow F_X(x) = (1-p)P(Y \leq x) + pP(Z \leq x)$$

$$\rightarrow = (1-p)F_Y(x) + pF_Z(x)$$

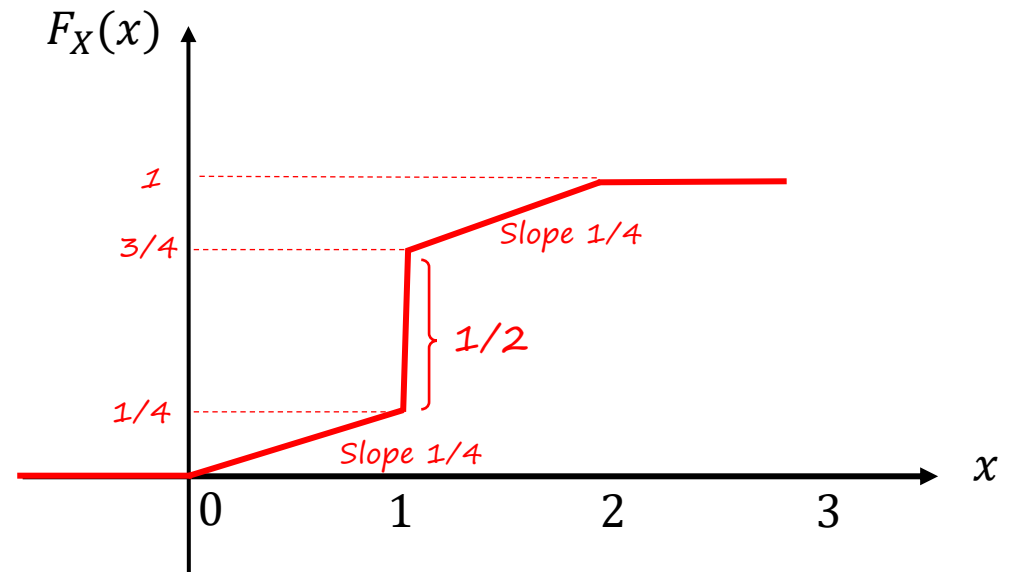
$$\rightarrow E[X] = (1-p)E[Y] + pE[Z]$$

Mixed Distributions

$$X = \begin{cases} 1, & \text{with probability } 1/2 \text{ } \mathbf{A2} \\ \text{uniform on } [0, 2], & \text{with probability } 1/2 \text{ } \mathbf{A1} \end{cases}$$



$$F_X(x) = \mathbf{P}(A_1) F_{X|A_1}(x) + \mathbf{P}(A_2) F_{X|A_2}(x)$$



Jointly continuous R.V.'s and Joint PDFs.

$$p_X(x) \quad f_X(x)$$

$$p_{X,Y}(x, y) \quad f_{X,Y}(x, y)$$

$$p_{X,Y}(x, y) = \mathbf{P} (X = x \text{ and } Y = y) \geq 0$$

$$f_{X,Y}(x, y) \geq 0$$

$$\mathbf{P}((X, Y) \in B) = \sum_{(X, Y) \in B} \sum p_{X,Y}(x, y)$$

$$\mathbf{P}((X, Y) \in B) = \iint_{(X, Y) \in B} f_{X,Y}(x, y) \, dx dy$$

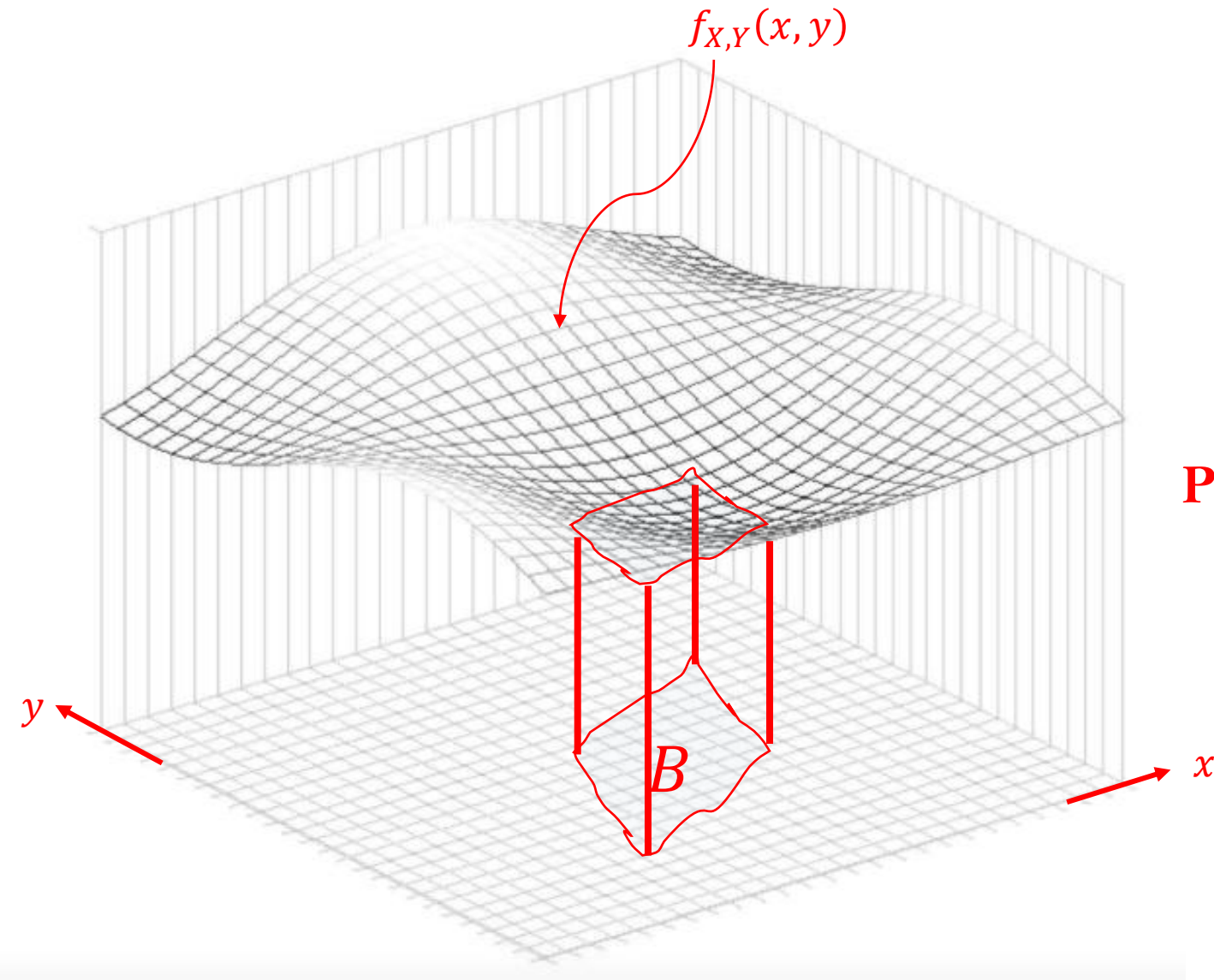
$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx dy = 1$$

Definition:

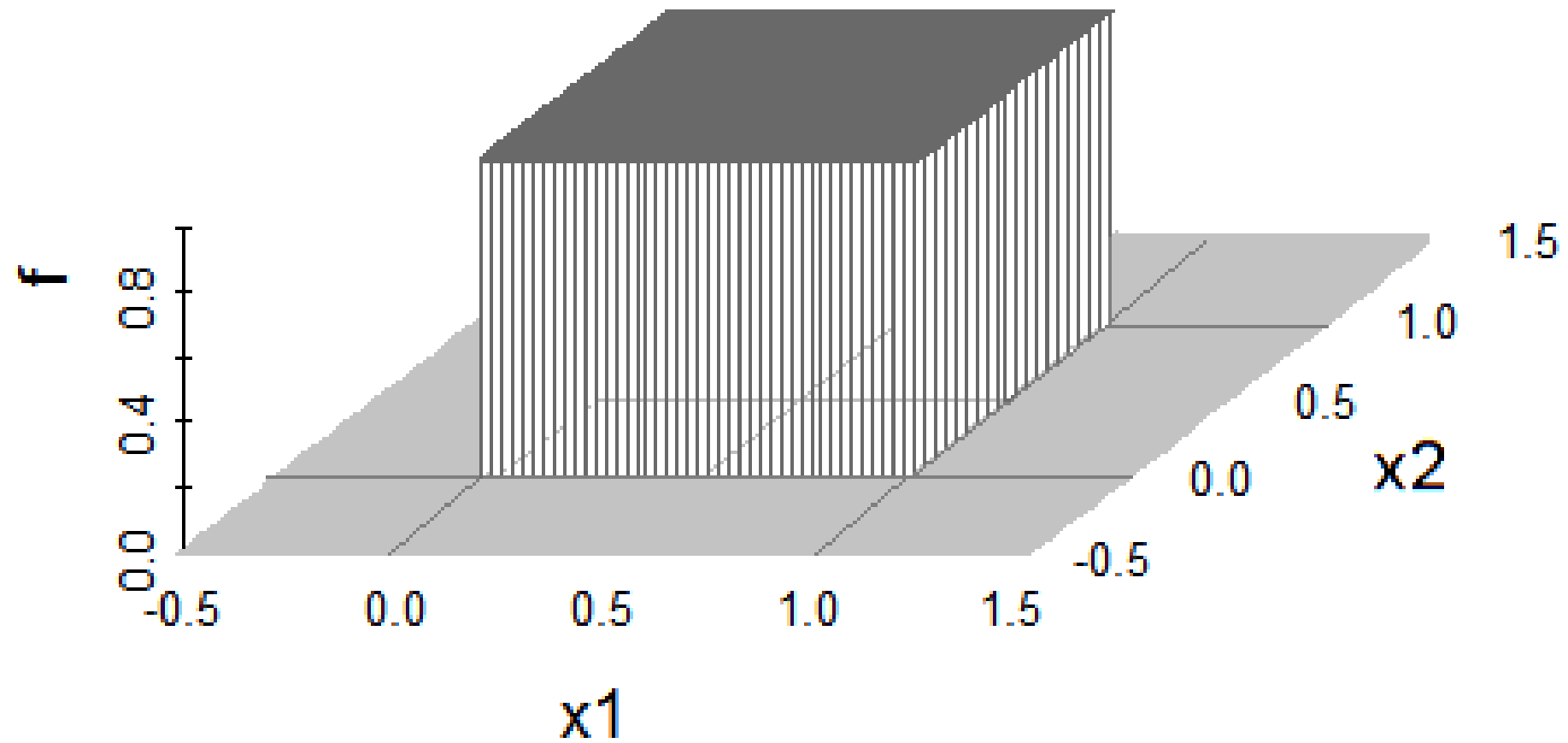
Two random variables are **jointly continuous** if they can be described by a joint PDF

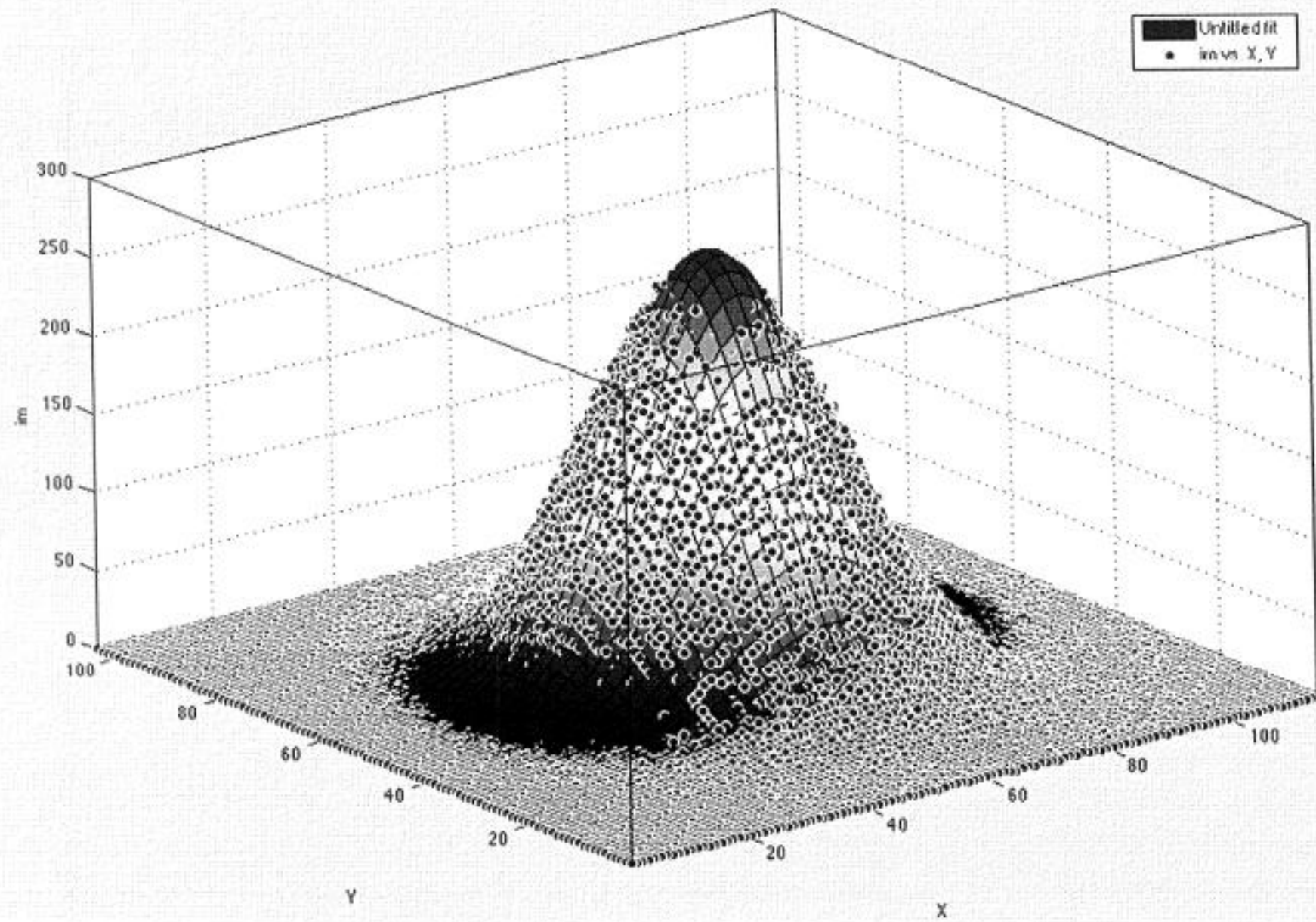
Visualizing a Joint PDF



$$\mathbf{P}((X, Y) \in B) = \iint_{(X, Y) \in B} f_{X,Y}(x, y) \, dx \, dy$$

bivariate uniform





On Joint PDFs

$$\mathbf{P}((X,Y) \in B) = \iint_{(X,Y) \in B} f_{X,Y}(x,y) dx dy$$

$$\mathbf{P}(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x,y) dx dy$$

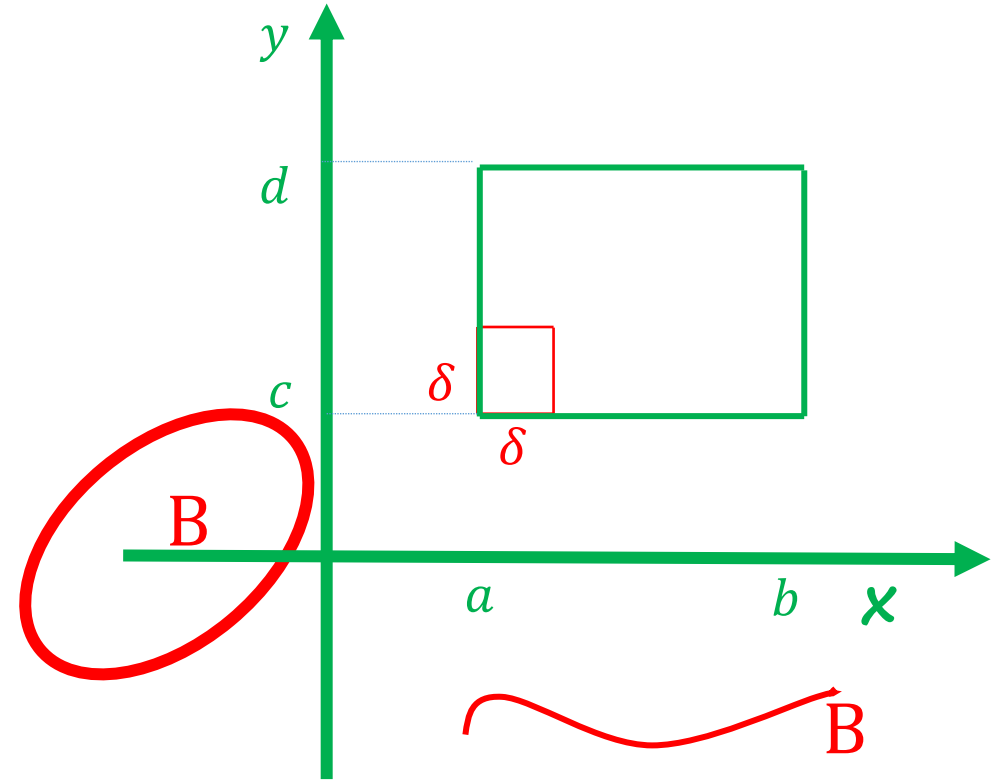
$$\mathbf{P}(a \leq X \leq a + \delta, c \leq Y \leq c + \delta) \approx f_{X,Y}(a, c) \delta^2$$

$f_{X,Y}(x,y)$: probability per unit area
around the point (x,y)

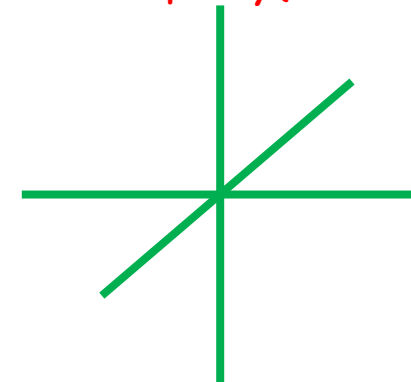
$$\text{area}(B) = 0 \Rightarrow \mathbf{P}((X,Y) \in B) = 0$$

Joint PDF

area

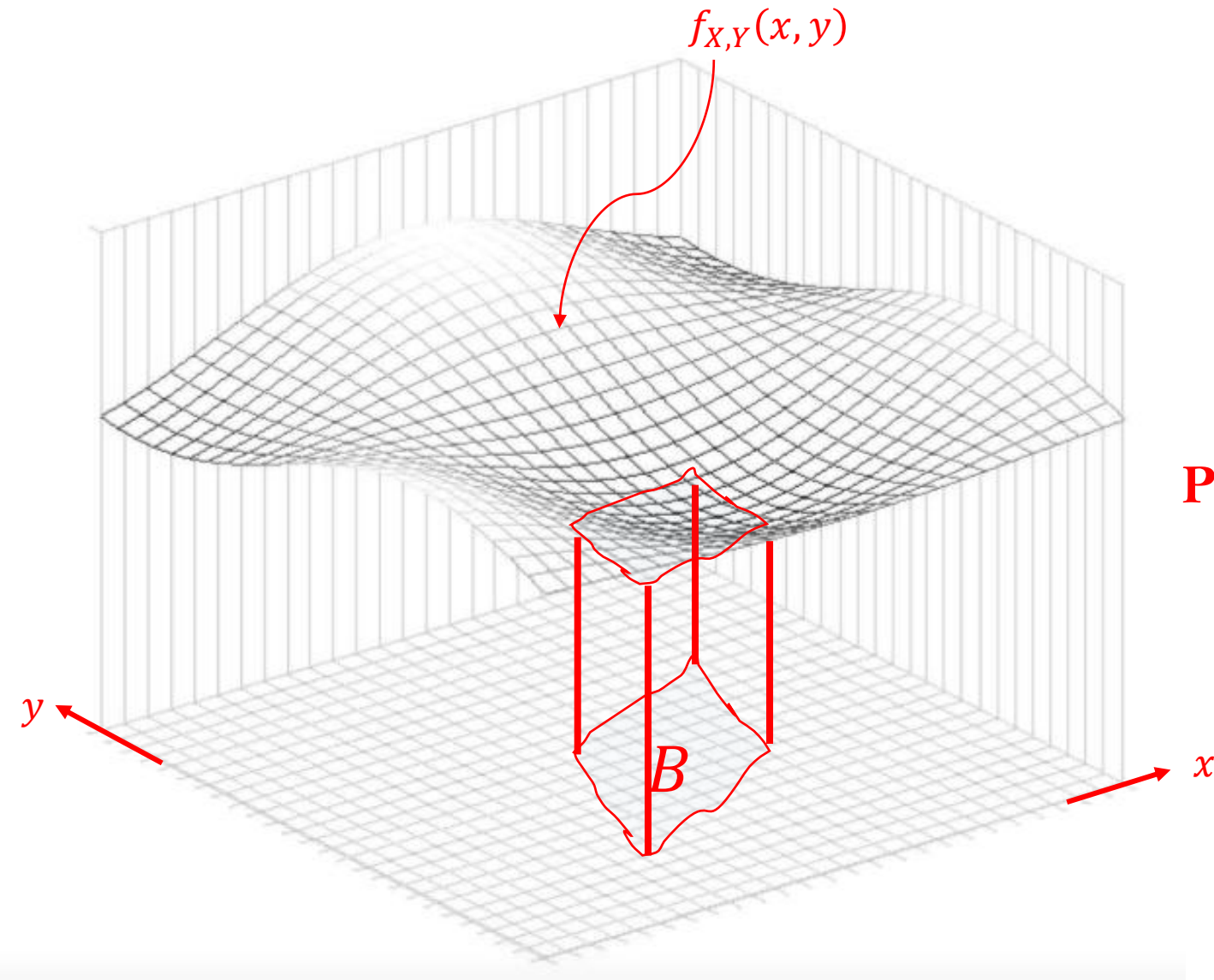


$Y = X$



Are X and Y jointly
continuous?

Visualizing a Joint PDF



$$\mathbf{P}((X,Y) \in B) = \iint_{(X,Y) \in B} f_{X,Y}(x,y) \, dx \, dy$$